Task 1 – Database Design (6 Marks)

- Design a complete database for the clinic that includes at least six entities such as Patients, Doctors, Appointments, Treatments, Medicines, Payments, or any other appropriate entities.
- A complete ER/EER diagram.
- Primary and foreign keys.
- Appropriate relationships with cardinality.
- At least one specialization/generalization (EER feature).
- Any assumptions made during the design process.

Assumptions:

- As we are creating a database for a private clinic, so we will need a payment entity.
- Doctor considered as a staff (staff is the superclass entity).
- Each appointment generates exactly one payment bill, and one treatment record.

First: Database Schema:

Entity	Attributes
Patient	PatientID, Name, DateOfBirth, ContactDetails, Address
Staff	StaffID, Name, DateOfBirth, ContactDetails, Address
Doctor	Specialty
Appointment	ApptID, ApptDate, ApptTime, Duration
Treatment	TreatmentID, Details
Medicine	MedicineID, Name, Stock
Payment	PaymentID, PayDate, PayTime, Amount

Second: Relationships & Cardinalities:

Relationship	Entities	Cardinality	Description
Is-A Relationship (Disjoint & Total Participation)	Staff – Doctor, Nurse, Receptionist	Inheritance / Specialization hierarchy	The staff must either a doctor, nurse or receptionist, but not more than one.
Booking	Patient - Appointment	1 : N	A patient can book many appointments, but the appointment is booked for only one patient.
Undertake	Doctor - Appointment	1 : N	A doctor can have many appointments, but each appointment is undertaken by one doctor.
Produce	Appointment - Treatment	1 : 1	Each appointment produces one treatment, and each treatment is linked with one appointment.
Include	Treatment - Medicine	M : N	The treatment could include many medicines, and the medicine could be in many treatments.
Incur	Appointment - Payment	1 : 1	Each appointment incurs one payment bill, and each bill is attached to one appointment.

Third: Relational Schema Mapping (Keys, Foreign Keys, Junction Table):

Entity	Attributes (Keys & Foreign Keys)
Patient	PatientID **(PK)**, Name, DateOfBirth , ContactDetails, Address
Staff	StaffID **(PK)**, Name, DateOfBirth , ContactDetails, Address
Doctor	StaffID **(PK, FK)**, Specialty
Appointment	ApptID **(PK)**, ApptDate, ApptTime, Duration, PatientID **(FK)**, Doctor.StaffID **(FK)**
Treatment	TreatmentID **(PK)**, Details, ApptID **(FK)**
Medicine	MedicineID **(PK)**, Name, Stock
Payment	PaymentID **(PK)**, PayDate, PayTime, Amount, ApptID **(FK)**
Treatment_ Medicine	TreatmentID **(PK, FK)**, MedicineID **(PK, FK)**, Quantity, Dosage **Composite PK: TreatmentID, MedicineID**

What to do next? Design the EER diagram.